

Mathematical Methods and Analysis in Asteroid Studies

Mithilesh Kumar and Upasana Rai

September 2026

Abstract

An asteroid usually appears as an unresolved point of light, so a telescope cannot directly show whether separate parts of its surface have different colours. Nevertheless, rotation repeatedly brings the same surface regions into and out of view. This paper examines how that repetition can be converted into statistical evidence using sparse observations obtained through different photometric filters. Its purpose is to reconstruct and explain, in accessible mathematical form, the surface-colour test proposed by Humes and Agarwal (2026), rather than to claim a new observational detection. The analysis begins with a uniformly coloured asteroid as the null model. After ordinary changes caused by distance, phase angle, shape, and rotation have been predicted, the remaining observed-minus-model differences are related to the surface region favoured by the Sun–asteroid–observer geometry. Regression determines the body-fixed direction in which those residuals vary most strongly. Rank correlations are then calculated independently for each filter, and their difference is converted into a significance statistic. Monte Carlo trials provide the probability that a uniform asteroid would generate an equally strong result by chance and reveal how the test responds to noise and incorrect model parameters. The central result of this analysis is that colour heterogeneity is not inferred from a single brightness or colour difference. It is inferred from a repeatable, location-dependent residual pattern whose strength changes with wavelength. Published detection-rate curves are used to show why adequate sampling and an accurate phase function, period, pole, and shape model are necessary before such a pattern can be treated as a credible candidate signal.

1. Introduction

1.1 Asteroids as unresolved rotating bodies

An asteroid is a small natural body orbiting the Sun. Most are composed mainly of rock, metal, carbon-rich material, or mixtures of these materials. Many are too small for gravity to pull them into a sphere, so their shapes are irregular. Most known asteroids lie in the main belt between Mars and Jupiter. Near-Earth asteroids approach Earth’s orbital region, while trans-Neptunian objects orbit farther from the Sun.

Asteroids preserve information about the formation and evolution of the Solar System. Their surfaces are changed by impacts, radiation, temperature cycling, movement of loose material, volatile loss, and space weathering. Consequently, two regions of one asteroid need not reflect light in exactly the same way. A colour difference can arise from different composition, grain properties, or exposure age. Measuring the frequency and strength of such differences would help constrain how frequently fresh material is exposed and how rapidly the space environment changes it.

For most asteroids this cannot be tested by simply taking a photograph. Their apparent angular size is much smaller than the resolution of a survey telescope. The detector records the light integrated over the entire visible part of the body. Surface information must therefore be inferred from how that total light changes as the object rotates and as the observing geometry changes.

1.2 Orbit and rotational state

An asteroid moves around the Sun along an approximately elliptical orbit. Six quantities commonly describe it:

Symbol	Physical meaning
a	Semimajor axis: characteristic orbit size.
e	Eccentricity: how stretched the ellipse is.
i	Inclination: tilt relative to the ecliptic.
Ω	Longitude of the ascending node.
ω	Direction of perihelion in the orbital plane.
M	Position along the orbit at a reference time.

An *ephemeris* is a time-indexed prediction of where an astronomical object will appear and of the geometry under which it is observed. Starting from the orbital elements and a reference epoch, an orbit propagator calculates the asteroid’s position and velocity at requested times. These Cartesian states are then converted into observer-centred quantities. A typical ephemeris row contains the UTC or Julian date, right ascension and declination (or altitude and azimuth), heliocentric distance r , observer distance δ , solar phase angle α , and solar elongation. Thus, an ephemeris does not provide an image; it supplies the predicted position and viewing geometry needed to interpret an image. JPL Horizons and the Minor Planet Center provide such information in web tables, text files, and machine-readable services [9, 10].

Orbit and rotation are different. Orbit describes travel around the Sun. Rotation period describes how quickly the body spins; the pole gives its spin-axis direction; and a shape model describes its three-dimensional form.

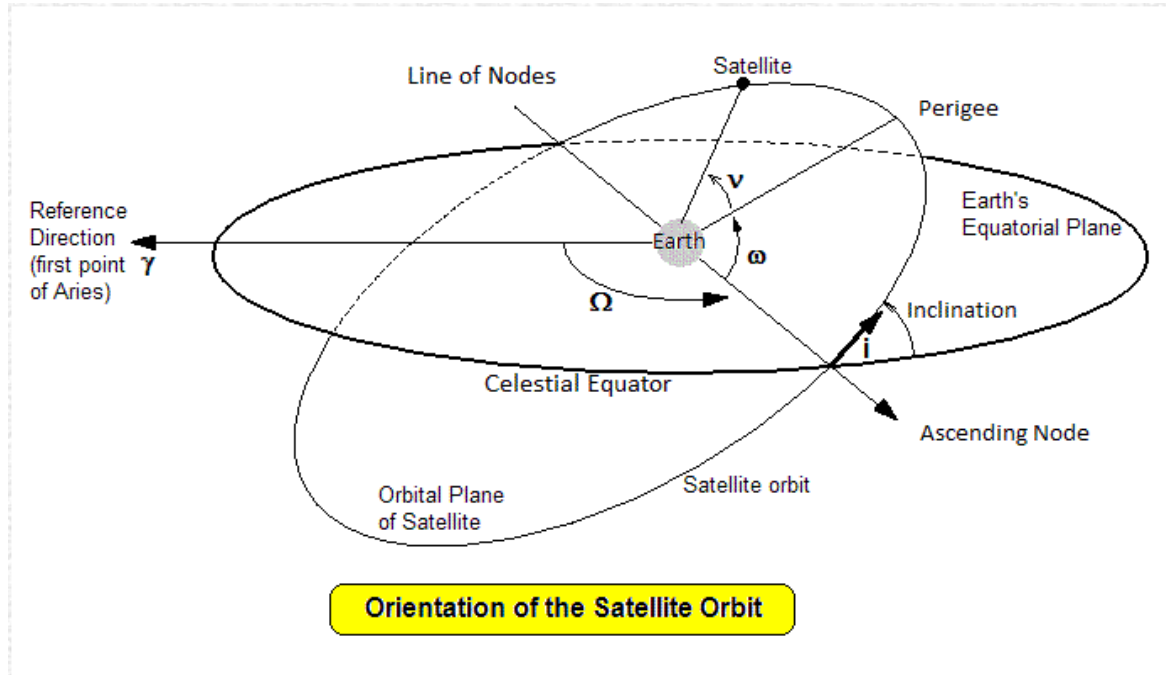


Figure 1: Schematic interpretation of the orbital elements. The semimajor axis a fixes the orbit's scale, while eccentricity $e = c/a$ measures the displacement c of the Sun from the ellipse centre relative to a . Within the orbital plane, ω sets the perihelion direction and M specifies the asteroid's position at a chosen epoch. The inclination i tilts the orbital plane relative to the reference ecliptic plane, and Ω locates the ascending node around that reference plane. The drawing is explanatory and not to scale.

1.3 Photometric observations

A survey telescope records an electronic image. Most asteroids are too small and distant to show a visible surface; they appear as points or short trails. Image processing measures position and received light, or *flux*. Results may be stored as FITS images and catalogue rows. The main fields are:

- **Time:** the exposure start, midpoint, or end, normally recorded in UTC, Julian Date (JD), or Modified Julian Date (MJD).
- **Filter:** the wavelength band transmitted by the instrument, for example ZTF g or r .
- **Magnitude:** the calibrated logarithmic brightness in magnitudes (mag).
- **Uncertainty:** the estimated 1σ measurement error, usually in magnitudes.
- **Coordinates:** sky position, commonly right ascension and declination in degrees, and sometimes image coordinates (x, y) in pixels.
- **Quality flags:** integer or bit-mask indicators of possible saturation, blending, detector defects, poor calibration, or other unreliable conditions.

The apparent magnitude in one band is defined relative to a reference flux F_0 :

$$m = -2.5 \log_{10} \left(\frac{F}{F_0} \right). \quad (1)$$

Here m is a dimensionless logarithmic quantity conventionally reported in magnitudes (mag). The measured flux F and reference flux F_0 must have identical units. Depending on the calibration stage, these may be detector count rate (electrons s^{-1} or ADU s^{-1}), flux density (Jy), or physical irradiance (W m^{-2}); only their ratio enters the logarithm.

For two measurements in the same photometric system,

$$\begin{aligned} m_1 &= -2.5 \log_{10}(F_1/F_0), \\ m_2 &= -2.5 \log_{10}(F_2/F_0). \end{aligned}$$

Subtracting the second equation from the first cancels the common zero point F_0 and gives

$$m_1 - m_2 = -2.5 \log_{10} \left(\frac{F_1}{F_2} \right). \quad (2)$$

Symbols. m_1, m_2 : apparent magnitudes in mag; F_1, F_2 : measured fluxes in the same physical or instrumental units; F_0 : reference or zero-point flux; $\log_{10}$: base-10 logarithm.

Interpretation. Smaller magnitude means brighter. A five-magnitude difference represents a factor of 100 in flux.

Different filters measure different wavelength ranges. ZTF commonly observes in g and r bands and releases images, source catalogues, and light curves through IRSA [11]. ATLAS uses broad cyan and orange bands. Repeated measurements form a *light curve*.

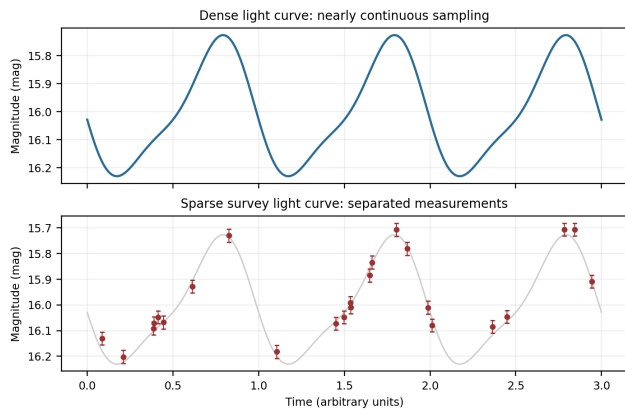


Figure 2: Illustrative dense and sparse sampling of the same rotating-asteroid light curve. Dense photometry traces the complete periodic variation, whereas a survey records separated points and may miss maxima or minima. The grey line in the lower panel is shown only to reveal the underlying curve; a real sparse dataset does not know that curve in advance. This schematic was generated for explanation and is not observational data.

Dense observations follow an object continuously; sparse survey photometry provides separated measurements over months or years.

1.4 Sources of brightness variation

Surface colour is only one cause of brightness variation. Brightness also changes with Sun and observer distance, solar phase angle, irregular shape, rotation, surface scattering, measurement noise, and calibration. A long axis turning toward the observer can brighten the asteroid even when every surface element has identical reflectance. Similarly, measurements obtained at larger phase angles are normally fainter. These ordinary effects must be modelled before an unexplained band-dependent pattern can be interpreted as possible surface heterogeneity.

The remainder of this paper first places the problem within earlier asteroid photometry research, then defines the uniform-colour hypothesis, develops the physical and statistical methodology equation by equation, and finally interprets the Monte Carlo experiments and published performance curves. The emphasis is on understanding where probability enters the analysis and what a statistically significant result can—and cannot—establish.

2. Background and previous research

Rotational photometry has long been used to infer periods and shapes. Convex light-curve inversion developed by Kaasalainen and collaborators estimates shape and spin from disk-integrated light [2, 3]. Hanuš et al. demonstrated large-scale use with sparse observations, while DAMIT stores published triangular shape and

spin solutions [4, 12].

Brightness and colour changes have revealed surface differences on Iapetus, Pluto, and Haumea. Lacerda’s multiband rotational study of Haumea reported evidence for a darker, redder region [5]. These studies establish the physical idea: as a body rotates, different regions contribute to its integrated light.

Modern studies have fitted phase curves and estimated average colours from sparse surveys. Mahlke et al. used ATLAS dual-band data for taxonomy [6]. Carry et al. treated rotation-aware survey analysis, and Oszkiewicz et al. used shape and viewing-geometry correction to improve ATLAS phase curves [7, 8].

Humes and Agarwal asked a different question: can sparse, non-contemporaneous multiband observations reveal that the *shape of rotational brightness variation* depends on wavelength? Their work develops the statistical test and evaluates it on synthetic asteroids; it does not present a catalogue of confirmed real detections [1].

3. Problem statement

A survey normally cannot resolve an asteroid’s surface. A difference between g and r gives average colour, but does not show whether the whole surface has that colour or one region is different. Shape, rotation, distance, and phase angle can also change brightness and may be sampled differently in each band.

The problem is:

Can the observations be distinguished from the brightness expected for a uniformly coloured asteroid, and is the unexplained pattern significantly different between wavelength bands?

The hypotheses are

- H_0 : uniform surface colour,
- H_1 : a region has different reflectance.

The test asks how unusual the data would be if H_0 were true. The scientific objective is not to fit any residual pattern, but to determine whether a pattern follows a fixed direction on the rotating asteroid and whether its strength is wavelength dependent. An unusual result makes the asteroid a candidate; it does not directly reveal composition or constitute final proof.

4. Methodology I: photometric null model

The physical model is included only to establish the expected light curve under H_0 .

4.1 Predicted magnitude

The predicted magnitude in band B is

$$m_B(t) = H_{0,B} + f_B[\alpha(t)] + 5 \log_{10}[r(t)\delta(t)] - 2.5 \log_{10} \left[\frac{F_B(t)}{F_{0,B}} \right]. \quad (3)$$

Symbols. t : time; B : filter; m_B : predicted magnitude; $H_{0,B}$: reference absolute magnitude; α : phase angle; f_B : band-dependent phase function; r : Sun–asteroid distance; δ : observer–asteroid distance; F_B : flux for the actual orientation; $F_{0,B}$: reference flux. Distances are in AU.

Interpretation. Start with reference brightness, then account for phase angle, distance, shape, and orientation. The result is what a uniform surface should produce.

The distance term arises because sunlight spreads before reaching the asteroid and reflected light spreads before reaching the observer. The flux ratio describes changing projected area during rotation.

4.2 Only the necessary surface geometry

The surface is represented by triangles. For facet T ,

$$\mu_{0,T}(t) = \hat{\mathbf{n}}_T \cdot \hat{\mathbf{E}}_0(t), \quad (4)$$

$$\mu_T(t) = \hat{\mathbf{n}}_T \cdot \hat{\mathbf{E}}(t). \quad (5)$$

Symbols. $\hat{\mathbf{n}}_T$: outward unit normal of triangle T ; $\hat{\mathbf{E}}_0$: unit vector toward the Sun; $\hat{\mathbf{E}}$: unit vector toward the observer; $\mu_{0,T}$: illumination cosine; μ_T : visibility cosine; $\cdot$: dot product.

If both cosines are positive, the facet faces the Sun and observer. Total flux is

$$F_B(t) = \sum_T A_T R_{T,B} S[\mu_T(t), \mu_{0,T}(t)]. \quad (6)$$

Symbols. A_T : facet area; $R_{T,B}$: reflectance; S : scattering function; $\sum_T$: sum over all facets.

Under H_0 , $R_{T,B}$ is constant over all facets in one band. This calculation removes ordinary rotational brightness; it is not itself the probability test.

The scattering term used in the paper combines Lommel–Seeliger and Lambert-like behaviour:

$$S = \mu_T \mu_{0,T} \left(\frac{1}{\mu_T + \mu_{0,T}} + 0.1 \right), \quad \mu_T, \mu_{0,T} > 0, \quad (7)$$

$$S = 0, \quad \text{otherwise.}$$

The zero case means that a triangle contributes no light if it faces away from either the Sun or the observer. The model assumes a convex body, so this sign test is sufficient and expensive ray tracing of self-shadowing is not required.

The timestamp should be corrected for light travel:

$$t_{\text{emit}} = t_{\text{receive}} - \frac{\delta}{c}. \quad (8)$$

Symbols. c : speed of light; δ/c : light-travel time.

Interpretation. Use the orientation when light left the asteroid, not when it arrived at Earth.

5. Methodology II: statistical test

The statistical analysis follows a connected sequence. First, the uniform-colour model predicts the asteroid's brightness at every observation time. Subtracting this prediction from the measured magnitude produces a residual. Each residual is then associated with the surface direction favoured by the illumination and viewing geometry. Correlations between residual magnitude and that direction are calculated separately for the different filters. Finally, the band correlations are compared to determine whether their difference is too large to be explained plausibly by a uniformly coloured asteroid.

5.1 Data and model inputs

For each observation the analysis requires time, photometric band, measured magnitude, and measurement uncertainty. An ephemeris supplies the Sun and observer directions and the distances r and δ . A rotational model supplies pole orientation, period, reference epoch, and initial rotation angle. A shape and scattering model then predicts brightness at the same time. Photometric parameters such as absolute magnitude and phase function may differ between filters, but physical shape and rotation must be common to all filters.

The null model is fitted under spatially uniform reflectance. This distinction is essential: the asteroid may have a different average brightness in g and r without being heterogeneous. The null hypothesis forbids only a change in colour from one surface location to another.

Figure 3 gives the complete logic visually. Shape and rotation cause both bands to vary in the upper light curve. Subtraction removes most of that common behaviour. The affected band retains repeated negative residuals whenever the enhanced-reflectance region contributes strongly. Geometry, rather than time alone, is therefore used to organise the remaining pattern.

5.2 Step 1: residuals

For every measurement,

$$\Delta m_B(t) = m_{B,\text{obs}}(t) - m_{B,\text{model}}(t). \quad (9)$$

Symbols. Δm_B : residual; $m_{B,\text{obs}}$: observed magnitude; $m_{B,\text{model}}$: uniform prediction.

Interpretation. The residual is what the uniform model did not explain. Negative means brighter than predicted; positive means fainter.

Conceptually,

$$\Delta m_B \approx \text{surface signal} + \text{noise} + \text{model error}. \quad (10)$$

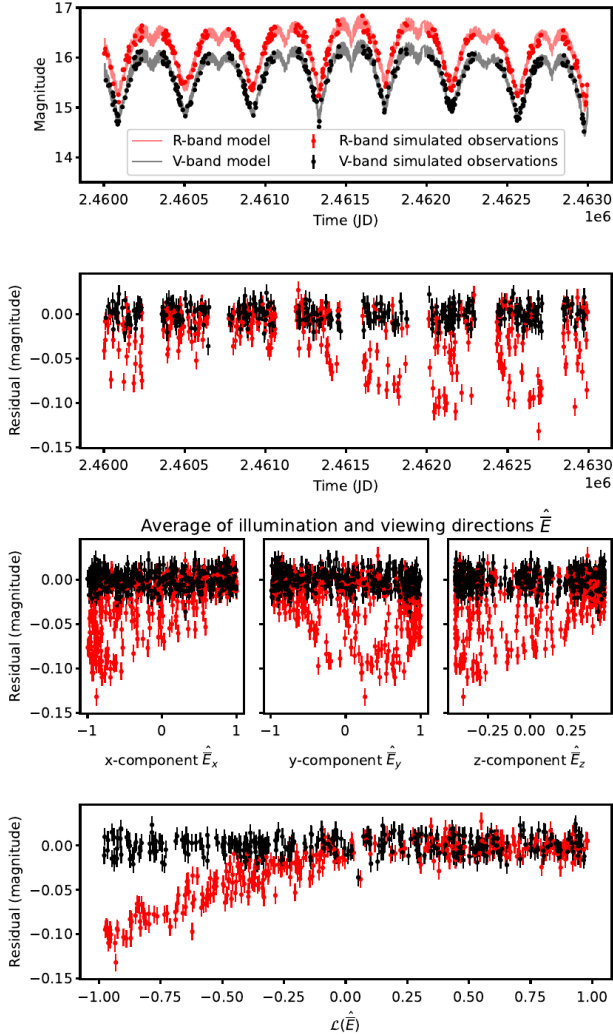


Figure 3: Published graphical summary of the statistical method. The top panel compares a uniform model with simulated observations in two bands. The second panel shows residual magnitude. The three small panels relate residuals to the components of the average illumination/viewing vector. The final panel projects that geometry onto the regression-selected direction $\hat{k}$. Red points contain the simulated wavelength-dependent spot; black points do not.

Source: Cropped from Humes and Agarwal (2026), Fig. 1 [1]. Original article distributed under the CC BY 4.0 licence.

One large residual is not a spot. A fixed feature should create a repeated pattern whenever its region is illuminated and visible.

The residual calculation must use the model evaluated at the exact observation epoch and in the corresponding filter. Fitting the phase curve separately in each filter prevents an ordinary difference in mean colour or phase reddening from being mistaken immediately for spatial heterogeneity.

5.3 Step 2: contributing direction

The authors combine the Sun and observer directions:

$$\hat{\mathbf{E}}(t) = \frac{1}{2}[\hat{\mathbf{E}}_0(t) + \hat{\mathbf{E}}(t)]. \quad (11)$$

Symbols. $\hat{\mathbf{E}}$: average geometry vector in the rotating asteroid frame; $\hat{\mathbf{E}}_0$: Sun direction; $\hat{\mathbf{E}}$: observer direction.

Interpretation. A region adds light only when lit and visible. The average vector approximately represents both conditions.

Its components $(\hat{\mathbf{E}}_x, \hat{\mathbf{E}}_y, \hat{\mathbf{E}}_z)$ rotate with the asteroid, so observations of the same surface region have related coordinates even on different dates.

The vector is not a precise location of the centre of measured light. It is a simple geometric proxy. Humes and Agarwal tested the Sun direction, observer direction, and their average, and found that the average gave better simulated detection because it accounts simultaneously for illumination and visibility.

5.4 Step 3: direction of greatest variation

A patch need not align with an axis. Multiple linear regression fits

$$\Delta m = k_x \hat{\mathbf{E}}_x + k_y \hat{\mathbf{E}}_y + k_z \hat{\mathbf{E}}_z + \epsilon. \quad (12)$$

Symbols. k_x, k_y, k_z : fitted weights; $\hat{\mathbf{E}}_x, \hat{\mathbf{E}}_y, \hat{\mathbf{E}}_z$: geometry components; ϵ : unexplained variation.

Define

$$\mathbf{k} = (k_x, k_y, k_z), \quad \hat{\mathbf{k}} = \frac{\mathbf{k}}{\|\mathbf{k}\|}. \quad (13)$$

Symbols. $\mathbf{k}$: coefficient vector; $\|\mathbf{k}\|$: its length; $\hat{\mathbf{k}}$: unit direction of maximum variation.

Then reduce three components to one predictor:

$$\mathcal{L}[\hat{\mathbf{E}}(t)] = \hat{\mathbf{E}}(t) \cdot \hat{\mathbf{k}}. \quad (14)$$

Symbols. $\mathcal{L}$: scalar geometry predictor; $\hat{\mathbf{k}}$: selected direction; $\cdot$: projection using the dot product.

Interpretation. Regression finds the direction that best organises unexplained brightness. $\mathcal{L}$ says how strongly each observation views that direction.

The direction is approximate, not a resolved surface map. Since it is selected from the residual data, this selection must be repeated inside Monte Carlo tests.

The regression is fitted using measurements from all bands together. This creates one common direction $\hat{\mathbf{k}}$ before the wavelength dependence is tested. If each band were allowed to choose an unrelated direction, random patterns would be easier to overfit and the two band correlations would not describe the same candidate surface location.

5.5 Step 4: Spearman correlation

For each band separately,

$$\rho_B = \text{corr}_{\text{Spearman}}(\mathcal{L}, \Delta m_B). \quad (15)$$

Symbols. ρ_B : Spearman correlation in band B ; $\mathcal{L}$: geometry predictor; Δm_B : residual.

Without tied ranks,

$$\rho_B = 1 - \frac{6 \sum_{i=1}^{n_B} d_i^2}{n_B(n_B^2 - 1)}. \quad (16)$$

Symbols. n_B : number of observations; d_i : difference between the ranks of $\mathcal{L}_i$ and $\Delta m_{B,i}$; i : observation index.

$\rho \approx +1$ means residual rises with $\mathcal{L}$; $\rho \approx -1$ means it falls; $\rho \approx 0$ means no monotonic pattern. Spearman correlation uses ranks, so the relationship need not be a perfect line.

The evidence is a *difference* between bands. A shape error may correlate similarly in both bands; a colour feature should affect their residual patterns differently.

Spearman correlation is used instead of an ordinary linear correlation because the relation between a visible fraction of a spot and projected geometry need only be monotonic. Ranking also reduces sensitivity to a few extreme residual values, although it does not remove the need for photometric quality control.

5.6 Step 5: Fisher transformation

Raw correlations are bounded by -1 and 1 . Fisher's transformation is

$$z_B = \text{arctanh}(\rho_B) = \frac{1}{2} \ln \left(\frac{1 + \rho_B}{1 - \rho_B} \right). \quad (17)$$

Symbols. z_B : transformed correlation; ρ_B : band correlation; $\ln$: natural logarithm; arctanh : inverse hyperbolic tangent.

For a sufficiently large independent sample,

$$\text{SE}(z_B) \approx \frac{1}{\sqrt{n_B - 3}}. \quad (18)$$

Symbols. SE: standard error; n_B : observations in the band.

Interpretation. The transformation puts correlations on a scale where their uncertainty is easier to compare with a normal approximation.

5.7 Step 6: compare two bands

For independently sampled bands B_1 and B_2 ,

$$z_{B_1, B_2} = \frac{z_{B_1} - z_{B_2}}{\sqrt{\frac{1}{n_{B_1} - 3} + \frac{1}{n_{B_2} - 3}}}. \quad (19)$$

Symbols. z_{B_1, B_2} : significance of the correlation difference; z_{B_1}, z_{B_2} : transformed correlations; n_{B_1}, n_{B_2} : band sample sizes.

The numerator is the measured difference. The denominator is its expected random uncertainty. Humes and Agarwal use $z_{B_1, B_2} > 5$ in their defined band ordering. If either band may contain the anomaly, sign handling or $|z|$ must be defined consistently and calibrated.

Equation (19) assumes that the two correlation estimates arise from independent samples. The paper applies it to non-contemporaneous measurements obtained at different epochs. If both filters are observed simultaneously, the samples are paired and direct colour, such as $g - r$, can instead be analysed as a function of geometry.

5.8 Step 7: probabilistic interpretation

Under an ideal normal approximation, the one-sided tail probability is

$$p = 1 - \Phi(z_{B_1, B_2}). \quad (20)$$

Symbols. p : p-value; Φ : standard-normal cumulative distribution; z_{B_1, B_2} : test statistic.

Interpretation. The p-value is how often a statistic this extreme would occur if the null model and assumptions were correct. It is not the probability that the asteroid is uniform; $1 - p$ is not automatically the probability of a colour patch.

Humes and Agarwal report the result primarily as the significance statistic in Eq. (19) and adopt a threshold of five standard deviations. Equation (20) expresses the corresponding idealised tail probability; it should not be mistaken for a separately inferred physical probability that a surface feature exists.

5.9 Worked example

Let $n_g = n_r = 100$, $\rho_g = -0.10$, and $\rho_r = -0.70$. Then

$$\begin{aligned} z_g &= \text{arctanh}(-0.10) \approx -0.100, \\ z_r &= \text{arctanh}(-0.70) \approx -0.867. \end{aligned}$$

The difference uncertainty is $\sqrt{1/97 + 1/97} \approx .1436$, hence

$$z_{g,r} = \frac{-0.100 - (-0.867)}{.1436} \approx 5.34. \quad (21)$$

The correlations differ by 5.34 estimated standard errors and pass the paper's 5σ rule in this ordering. The number is not a percentage and does not measure spot size.

6. Monte Carlo analysis

6.1 Why simulation is necessary

The Fisher formula assumes independent samples and a fixed analysis. Here $\hat{\mathbf{k}}$ is chosen using the same residuals being tested. Real times, unequal band coverage, uncertainties, and model errors can also change the distribution. The complete procedure is therefore repeated on synthetic datasets with known truth.

Random magnitude noise is represented by

$$m_{\text{sim}}(t) = m_{\text{truth}}(t) + \eta(t), \quad \eta \sim \mathcal{N}(0, \sigma_m^2). \quad (22)$$

Symbols. m_{sim} : simulated measurement; m_{truth} : exact value; η : random error; $\mathcal{N}(0, \sigma_m^2)$: normal distribution with mean 0 and variance σ_m^2 ; σ_m : magnitude uncertainty.

Each trial generates observations, adds noise, applies the uniform model, calculates residuals, selects $\hat{\mathbf{k}}$, computes correlations, and obtains the final statistic. Repeating the trial for uniform and spotted surfaces reveals performance.

For a real-data implementation, an empirical null probability may be estimated as

$$\hat{p} = \frac{1 + \sum_{j=1}^{N_{\text{MC}}} \mathbf{1}(T_j \geq T_{\text{obs}})}{N_{\text{MC}} + 1}. \quad (23)$$

Symbols. $\hat{p}$: estimated p-value; N_{MC} : number of null trials; T_j : statistic from trial j ; T_{obs} : measured statistic; $\mathbf{1}$: 1 when the condition is true, otherwise 0.

Interpretation. Count uniform simulations that look at least as extreme as the data. Few such trials mean the data are unusual under the uniform model.

This empirical expression makes explicit how a Monte Carlo calibration could assign a tail probability to one observed object. In the published paper, the simulations are used mainly to measure detection and false-positive rates across populations of synthetic asteroids, rather than to publish an object-by-object posterior probability.

6.2 Detection and false positives

$$\text{TPR} = \frac{N_{\text{detected,spot}}}{N_{\text{spot}}}, \quad (24)$$

$$\text{FPR} = \frac{N_{\text{detected,uniform}}}{N_{\text{uniform}}}. \quad (25)$$

Symbols. TPR: true-positive or detection rate; FPR: false-positive rate; numerator: number passing the threshold; denominator: number tested in that class.

High TPR means simulated spots are usually found. Low FPR means uniform objects are rarely flagged. Both are required; a method that flags everything has high TPR but no scientific value.

6.3 Simulation design reported by Humes and Agarwal

The paper drew 1000 examples each from main-belt, near-Earth, and trans-Neptunian orbit populations with $H < 15$. Orbital distributions were sampled from JPL data and propagated with *kete*. For each band, 300 observation dates were drawn over 3000 days; epochs with solar elongation below 45° were rejected to imitate a ground-based survey constraint. Spin poles were drawn uniformly on a sphere, initial rotation angle was random, and most periods were drawn between 0.1 and 2 days.

The simulated shapes were triangulated asymmetric ellipsoids. Phase functions were allowed to depend on wavelength. For every asteroid the authors created two datasets: a uniform surface with constant reflectance and a spotted surface with enhanced reflectance in one of the two bands. A spot was produced by selecting facets within an angular radius ϕ_{spot} of a randomly selected surface direction. Spot radii ranged from about 20° to 90° , and the reflectance enhancement R_f ranged from 1.2 to 2.0. Baseline Gaussian magnitude noise had standard deviation 0.1 mag [1].

To examine systematic errors, the true model parameters were deliberately perturbed before the uniform model was evaluated. The study changed period, pole orientation, shape, phase function, and related parameters, repeating the statistical test to measure how model error changes both TPR and FPR. This is an important methodological point: a probability method should be tested not only against random noise but also against plausible incorrect physical models.

This answers a controlled question: if a feature of a particular strength existed and the model had a particular error, how often would the procedure find it?

7. Results and interpretation

The simulations produced four main results:

1. Larger spots and stronger colour contrast were detected more frequently.
2. More observations improved detection. Under tested conditions, roughly 100–150 total measurements often produced about 90% detection, depending on spot properties.
3. Increasing photometric noise reduced detection. Random noise more often hid a true pattern than created a stable one.
4. Rotation-period error most strongly reduced true detections; phase-function error most strongly raised false positives.

Ideal baseline false-positive rates were generally at or below about 0.1%. Main-belt detection was strongest and TNO detection lower because TNO viewing geometry changes slowly. Both bands need enough observations and similar geometry coverage;

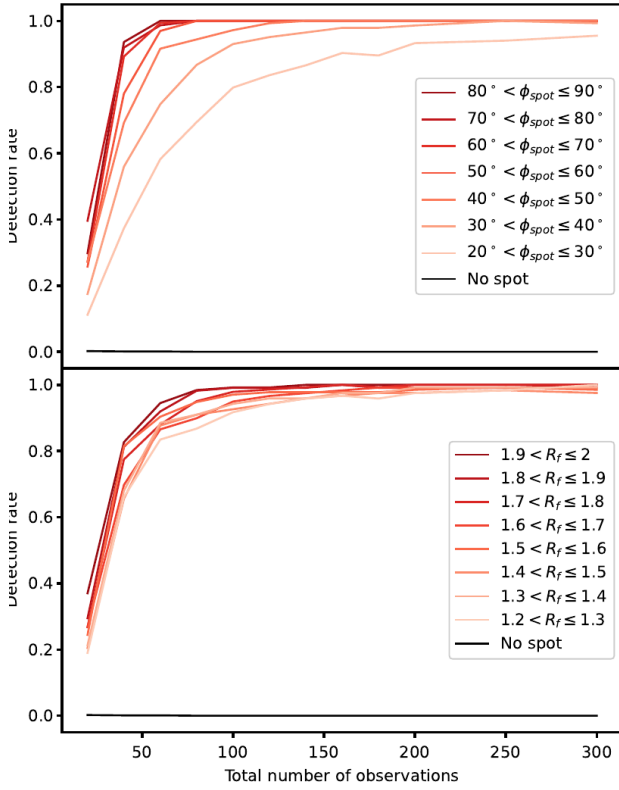


Figure 4: Detection rate versus total number of observations. The upper panel groups simulations by spot angular size ϕ_{spot} ; the lower groups them by reflectance enhancement R_f . Detection approaches unity as more measurements are supplied. Larger and higher-contrast regions require fewer measurements. The black no-spot curve remains near zero.

Source: Cropped from Humes and Agarwal (2026), Fig. 3 [1]. Original article distributed under the CC BY 4.0 licence.

strong imbalance makes their correlations unfair to compare.

Figure 4 explains why “enough observations” cannot be expressed as one universal number. A large, high-contrast region becomes detectable quickly, whereas a small, weak feature can remain undetected even with a useful dataset. Observation count must therefore be interpreted together with geometry coverage, precision, and the type of feature the experiment could recover.

Figure 5 shows that random error mainly destroys a real correlation rather than creating one. This should not be confused with systematic error. A wrong phase function can leave an ordered residual pattern and can therefore raise false positives even when random noise alone does not.

A positive result supports only this statement:

The band-dependent residual pattern is difficult to produce using the tested uniformly coloured model.

It does not prove a mineral, produce a surface image, or

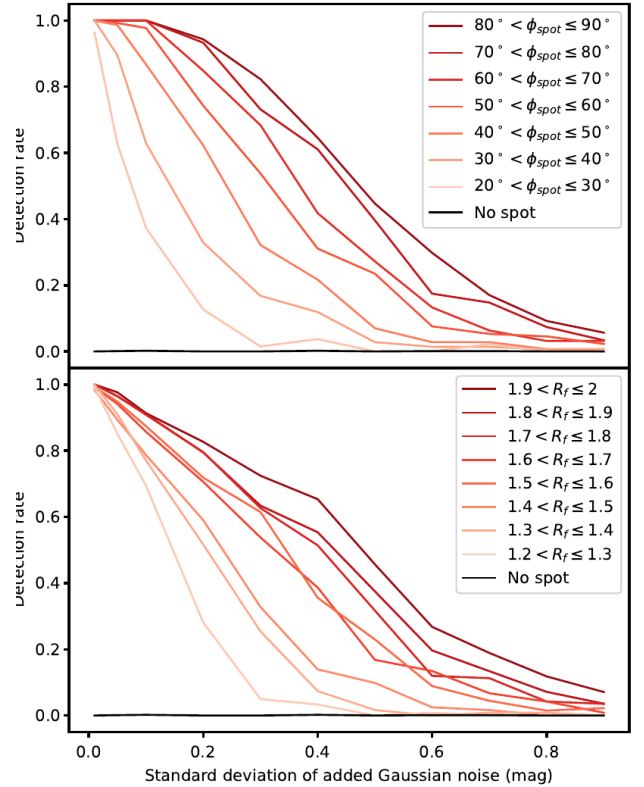


Figure 5: Detection rate as Gaussian magnitude noise increases. The upper panel groups spot sizes and the lower panel reflectance contrasts. Detection falls first for small and low-contrast regions. The black no-spot curve remains close to zero in this ideal random-noise experiment.

Source: Cropped from Humes and Agarwal (2026), Fig. 5 [1]. Original article distributed under the CC BY 4.0 licence.

guarantee the physical model is correct. Wrong period, phase function, shape, calibration, contamination, or unequal sampling may imitate a signal.

8. Conclusion

The method begins with physical modelling and ends with probability. Residuals show what a uniform model fails to explain. The average Sun–observer vector relates that failure to body-fixed geometry. Regression selects the dominant direction. Spearman correlation measures the pattern in each band, Fisher’s transformation compares the correlations, and Monte Carlo experiments determine how unusual the result is and measure detection and false-positive rates.

The central lesson is that probability is conditional. A small p-value means “unusual under the assumed uniform model,” not “certainly a colour patch.” Reliability depends on the observations and on the rotation and phase models.

Equation map

Equation	Purpose
(3)	Predict uniform brightness.
(9)	Calculate unexplained magnitude.
(11)	Represent contributing direction.
(12)–(14)	Find maximum-variation direction.
(15)	Measure relation in each band.
(17)–(19)	Compare correlations.
(20), (23)	Calculate null probability.
(25)	Measure detections and false alarms.

References

- [1] Humes, O., & Agarwal, J. (2026). Prospects for detecting surface color heterogeneity on asteroid surfaces from sparse multiband photometric survey data. *A&A*, 711, A209. doi:10.1051/0004-6361/202659422.
- [2] Kaasalainen, M., & Torppa, J. (2001). Optimization methods for asteroid lightcurve inversion I. *Icarus*, 153, 24–36.
- [3] Kaasalainen, M., Torppa, J., & Muinonen, K. (2001). Optimization methods for asteroid lightcurve inversion II. *Icarus*, 153, 37–51.
- [4] Hanuš, J., Durech, J., Brož, M., et al. (2011). Asteroid shape models from light-curve inversion. *A&A*, 530, A134.
- [5] Lacerda, P. (2009). Time-resolved near-infrared photometry of Haumea. *AJ*, 137, 3404–3413.
- [6] Mahlke, M., Carry, B., & Denneau, L. (2021). Asteroid taxonomy from ATLAS dual-band photometry. *Icarus*, 354, 114094.
- [7] Carry, B., et al. (2024). Sparse survey photometry of asteroids. *A&A*, 687, A38.
- [8] Oszkiewicz, D., et al. (2026). Asteroid phase curves corrected for shape and geometry. *Icarus*, 446, 116886.
- [9] NASA/JPL Solar System Dynamics. Small-Body Elements and Horizons. https://ssd.jpl.nasa.gov/sb/elem_tables.html.
- [10] Minor Planet Center. Optical Observation Format. <https://minorplanetcenter.net/iau/info/OpticalObs.html>.
- [11] Zwicky Transient Facility. Public Data Releases. <https://www.ztf.caltech.edu/ztf-public-releases.html>.
- [12] Database of Asteroid Models from Inversion Techniques. <https://damit.cuni.cz/>.